

Malyala Sahithi Purna

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EDUCATION

CRRao AIMSCS

Hyderabad, Telangana

B.Tech in Computer Science & Engineering | SGPA: 9.1/10 Expected May 2028

EXPERIENCE

iHub, IIIT Hyderabad | Research Intern

March 2026 – May 2026

Weapon Brandishing Detection using Pose Estimation & Segmentation | Python, Ultralytics YOLO, OpenCV

- ▶ Built a custom YOLO segmentation pipeline to detect and segment firearms and bladed weapons in surveillance-style images; integrated pretrained pose estimation models to extract human keypoints for weapon-interaction analysis.
- ▶ Identified and documented a key failure mode — pose estimation consistently predicted 16–17 keypoints even on partially visible humans, making wrist-weapon spatial inference unreliable; findings were recorded and the project concluded with a structured analysis of the limitation.

TECHNICAL SKILLS

Languages: Python, C

Frameworks & Libraries: Pandas, NumPy, Seaborn, Scikit-learn, Matplotlib, SciPy, Ultralytics

Tools & Platforms: GitHub, VS Code

PROJECTS

Dimensionality Reduction on HAR Dataset | Python, Scikit-learn, Pandas

March 2026

- ▶ Benchmarked multiple classification and dimensionality reduction methods on the HAR dataset (561 features, 6 human activity classes from smartphone sensor data), systematically comparing trade-offs between model performance and computational cost.
- ▶ Evaluated each method against hard constraints — inference time, F1 score, and accuracy — producing a structured performance analysis to guide practical model selection under real-world resource constraints.

ML Projects — STP in AI/ML, IIIT Hyderabad | Python, Scikit-learn, NumPy, Pandas

Sep 2025 – March 2026

- ▶ Built and evaluated supervised and unsupervised models across multiple datasets, covering the full ML pipeline: data cleaning, feature engineering, model training, and performance evaluation.
- ▶ Worked on diverse problem settings as part of the structured training programme, gaining hands-on experience with classification, regression, and clustering across varied real-world datasets.

EDA & Hypothesis Testing — Statistical Inference | SciPy, NumPy, Seaborn, Scikit-learn, Matplotlib

December 2025

- ▶ Conducted multi-stage Exploratory Data Analysis — distribution analysis, correlation heatmaps, outlier detection — and applied PCA and t-SNE to visualize cluster separability and evaluate variance retention.
- ▶ Built a statistical inference pipeline covering confidence intervals, t-tests, chi-square, and ANOVA; documented all findings with annotated visualizations in a reproducible Colab Notebook.

ACHIEVEMENTS & EXTRACURRICULARS

- ▶ **HackElevate 2025, CRRao AIMSCS** — Pitched an AI-based Road Damage Detection proposal to a panel of judges, advancing to Semi-finals.
- ▶ **Crypto Conclave, CCS IITH** — Selected to attend a research-focused conclave on Cryptology, engaging with domain experts and researchers.
- ▶ **OSINT Club, Ex-Deputy** — Co-founded club from scratch; handled planning and early operations over a 2-month formation period.
- ▶ **CBITMUN, DISEC Committee** — Represented Finland; researched and debated disarmament agenda in the context of the Russia-Ukraine conflict.
- ▶ **Student Support Committee Head & CR- CSE Department** — Elected CR for CSE; extended coordination across all departments for college-level initiatives, while serving as liaison between students and management.
- ▶ **YuvaKala '25, Ethnic Day** — Scripted and planned the hosting flow for a college-wide cultural event; coordinated pre-event logistics and delivered live hosting on the day.

CERTIFICATIONS

Google AI Essentials — Google

October 2024

Student Training Programme in AI/ML — iHub, IIIT Hyderabad

March 2026